

Experimental Validation of Jacobian Bounds for Iterative Transformers

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Background

Akiyoshi Tomihari and Ryo Karakida (2025) analyze Multi-Head Self-Attention (MSA) as a dynamical system using Jacobian-based analysis.

Key findings:

- ▶ RMSNorm has a strong stabilizing effect on iterative attention
- ▶ Prevents gradient explosion in recurrent self-attention dynamics
- ▶ Introduced *Iterative Self-Attention (ItrSA)*:

$$\Delta Y^{(t)} = C + \text{MSA}(Y^{(t)}), \quad Y^{(t+1)} = \text{RMSNorm}(Y^{(t)} + \nu \Delta Y^{(t)})$$

Our contribution: extend this analysis to the full Transformer block (MSA + FFN + three RMSNorm layers, Post-Norm) and validate the theoretical bounds experimentally.

Problem Statement

Iterative Transformer block (Post-Norm) with state $Y^{(t)} \in \mathbb{R}^{S \times D}$ and context $C \in \mathbb{R}^{S \times D}$:

$$\begin{aligned} Y' &= \text{RMSNorm}\left(\underbrace{Y + \text{MSA}(Y)}_{Z_1}\right), \\ \text{TB}(Y) &= \text{RMSNorm}\left(\underbrace{Y' + \text{FFN}(Y')}_{Z_2}\right), \\ F(Y) &= \text{RMSNorm}\left(\underbrace{Y + \nu(C + \text{TB}(Y))}_{U(Y)}\right). \end{aligned}$$

Research objectives:

- ▶ Numerically validate the theoretical bound for $\|J_F(Y)\|_2$
- ▶ Study asymptotic behavior as $\nu \rightarrow \infty$
- ▶ Confirm the stabilizing role of RMSNorm
- ▶ Compare a hierarchy of tighter Jacobian bounds

Theoretical Foundation (Theorem 3.1)

Theorem 1 (Composed bound for the full block)

Suppose the row-wise lower bounds hold for all i :

- ▶ $\|Z_1[i, :]\| = \|(Y + \text{MSA}(Y))[i, :]\| \geq r_{\text{in}}$
- ▶ $\|Z_2[i, :]\| = \|(Y' + \text{FFN}(Y'))[i, :]\| \geq r_{\text{out}}$
- ▶ $\|U(Y)[i, :]\| = \|(Y + \nu(C + \text{TB}(Y)))[i, :]\| \geq r_U$

Let $\gamma_{\max} = \max_j |\gamma_j|$. Then

$$\|J_F(Y)\|_2 \leq \frac{\gamma_{\max}}{r_U} \left(1 + \nu \frac{\gamma_{\max}^2}{r_{\text{out}} r_{\text{in}}} (1 + \|J_{\text{FFN}}\|) (1 + \|J_{\text{MSA}}\|) \right).$$

Asymptotics (Theorem 3.2)

Under non-degeneracy $\min_i \|\Delta Y_{[i, :]}\|_2 \geq \varepsilon > 0$: $\|J_F(Y)\|_2 = O(1)$
as $\nu \rightarrow \infty$.

Experimental Methodology

Numerical estimation of the spectral norm:

- ▶ Power iteration with JVP (Jacobian–vector products)
- ▶ 20–50 iterations for convergence
- ▶ Automatic differentiation (PyTorch)

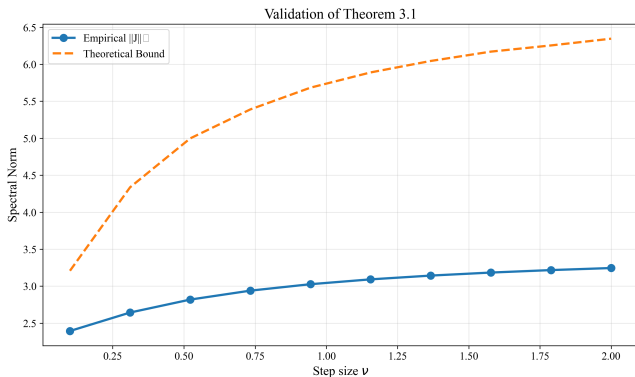
Data:

- ▶ Synthetic: $Y^{(0)}, C \sim \mathcal{N}(0, \sigma^2 I)$
- ▶ CIFAR-10: image patches as context

Model configurations:

- ▶ Exp. 1: $S = 32, D = 64, H = 4, D_{ff} = 128, \nu \in [0.1, 2.0]$
- ▶ Exp. 2: $S = 16, D = 32, H = 4, D_{ff} = 64, \nu \in [0.1, 100]$

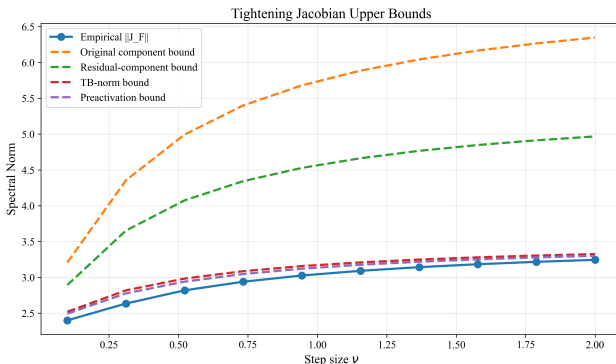
Experiment 1: Validation of the Theoretical Bound



Results:

- ▶ Empirical norm $\|J_F\|_2$ stays consistently below the bound
- ▶ Bound tracks the growth pattern in ν correctly
- ▶ Tightness factor: $\approx 2\times$ for $\nu \leq 2.0$

Hierarchy of Jacobian Bounds



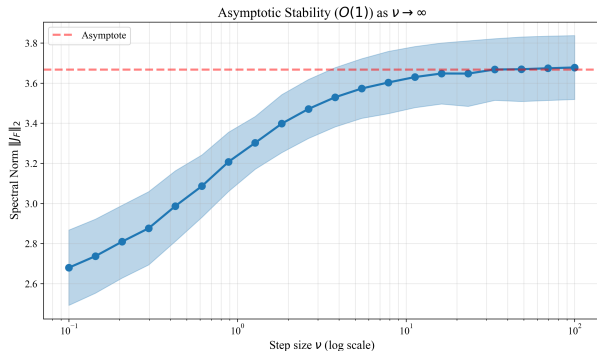
Curves on the plot (from loosest to tightest):

- ▶ **Original component bound** (Thm. 3.1):

$$\frac{\gamma_{\max}}{r_U} (1 + \nu \frac{\gamma_{\max}^2}{r_{\text{out}} r_{\text{in}}} (1 + \|J_{\text{FFN}}\|)(1 + \|J_{\text{MSA}}\|))$$

- ▶ **Residual-component bound**: replaces $(1 + \|J\|)$ with $\|I + J\|$ for MSA and FFN
- ▶ **TB-norm bound**: $\frac{\gamma_{\max}}{r_U} (1 + \nu \|J_{\text{TB}}(Y)\|_2)$ — avoids splitting MSA $\times$ FFN

Experiment 2: Asymptotic Behavior



Key observations:

- ▶ Sub-linear growth for $\nu \leq 1$
- ▶ Plateau reached at $\nu \approx 10$
- ▶ Stabilization at $\|J_F\|_2 \approx 3.7$ over four decades of ν

Compensation Mechanism

Theoretical explanation of $O(1)$ stability:

As $\nu \rightarrow \infty$, the preactivation $U(Y) = Y + \nu(C + \text{TB}(Y))$ is dominated by $\nu \Delta Y$:

$$r_U = \min_i \|U[i, :]\| \sim \nu \cdot \varepsilon$$

$$\|J_{\text{RMSNorm}}(U)\|_2 \leq \frac{\gamma_{\max}}{r_U} \sim \frac{1}{\nu}$$

Compensation

The inner block Jacobian contributes an $O(\nu)$ factor (chain rule), while the outer RMSNorm contributes $O(1/\nu)$.

Result: $\|J_F\|_2 \sim \frac{2\gamma_{\max}}{\varepsilon} = O(1)$ — independent of ν .

Conclusions

Main results:

1. **Theory validation:** empirical norms lie within a factor of 1.5–2.5 from the Thm. 3.1 bound.
2. **Asymptotic stability:** compensation $\gamma_{\max}/r_U \sim 1/\nu$ cancels the $O(\nu)$ growth $\Rightarrow \|J_F\|_2 = O(1)$.
3. **Bound hierarchy:** preactivation and TB-norm refinements reduce the gap to empirical from $\sim 2\times$ to almost $1\times$.
4. **Role of RMSNorm:** prevents exponential Jacobian growth even for large ν .
5. **Robustness:** confirmed on synthetic data and CIFAR-10 across various architectural configurations.

Practical significance

Iterative Transformers with RMSNorm provide controlled dynamics across many iterations — crucial for reasoning and looped architectures.